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# Repairing Crude Oil, Liquefied Petroleum Gas, and Product Pipelines

API RECOMMENDED PRACTICE 2200
THIRD EDITION, MAY 1994

**American Petroleum Institute**
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**Manufacturing, Distribution and Marketing Department**

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# FOREWORD

This recommended practice was prepared under the auspices of the API Pipeline Transportation Committee. It is intended for use by API member companies and others to develop safe practices for the repair of pipelines in crude oil, liquefied petroleum gas, and product service. The information presented represents a consensus on desirable approaches to repairs.

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# Repairing Crude Oil, Liquefied Petroleum Gas, and Product Pipelines

## SECTION 1—GENERAL

### 1.1 Scope

This recommended practice discusses guidelines to safe practices while repairing pipelines for crude oil, liquefied petroleum gas, and product service. Although it is recognized that the conditions of a particular job will necessitate an on-the-job approach, the observance of the suggestions in this document should improve the probability that repairs will be completed without accidents or injuries.

### 1.2 Conformance to API Guiding Environmental Principles

This recommended practice has been reviewed to determine its conformity to the API Guiding Environmental Principles.[1] The reviewers determined that this recommended practice directly addresses safety and environmental issues and all 11 principles. The reviewers further determined that the following petroleum industry guiding principles were especially relevant to this practice:

- To recognize and to respond to community concerns about our raw materials, products and operations.
- To operate our plants and facilities, and to handle our raw materials and products in a manner that protects the environment, and the safety and health of our employees and the public.
- To advise promptly appropriate officials, employees, customers and the public of information on significant industry-related safety, health and environmental hazards and to recommend protective measures.

The members of the American Petroleum Institute pledge to manage our businesses in accordance with these principles.

### 1.3 Referenced Publications

The following standards and codes are cited in this recommended practice:

API
- Std 1104 *Welding of Pipelines and Related Facilities*
- RP 1107 *Pipeline Maintenance Welding Practices*
- Publ 2209 *Pipe Plugging Practices* (Out of Print)

ASME[2]
- B31.4 *Liquid Transportation Systems for Hydrocarbons, Liquid, Petroleum Gas, Anhydrous Ammonia, and Alcohols*

DOT[3]
- 49 *Code of Federal Regulations* Part 195

OSHA[4]
- 29 *Code of Federal Regulations* Part 1910.120 and Part 1926, Subpart P

[1] *Charter and Bylaws of the American Petroleum Institute*, American Petrolum Institute, Washington, D.C., April 3, 1991.

[2] American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.

[3] Department of Transportation. The *Code of Federal Regulations* is available from the U.S. Government Printing Office, Washington, D.C. 20402.

[4] Occupational Safety and Health Administration, U.S. Department of Labor. The *Code of Federal Regulations* is available from the U.S. Government Printing Office, Washington, D.C. 20402.

## SECTION 2—PRELIMINARY KNOWLEDGE

### 2.1 Personnel

Qualified oversight is a prerequisite to the implementation of pipeline repair practices. If it is necessary for the designated supervisor or repair team leader to be absent, accountability should be specifically assigned to a responsible and experienced employee to act as a temporary supervisor. Personnel working on pipeline repairs must understand the importance of careful planning to ensure safe and efficient practices and procedures. Employees should be briefed onsite about specific repairs and safety procedures.

### 2.2 Safety Procedures

Personnel must have a basic knowledge of safety procedures that includes lockout and tagout, confined space, worker's right-to-know, personal protective equipment, excavation and trenching, fire prevention and protection, emergency response, and atmospheric testing devices.

The supervisor or repair team leader should review previous testing and repair files to determine the presence or absence of asbestos-containing pipe coatings, lead-based paint, or other substances requiring special attention. If coatings



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contain asbestos, special work practices designed by the company's health and safety department to mitigate employee exposure should be used.

In addition, the supervisor or repair team leader should be knowledgeable about the most recent Department of Transportation Alert Notices and National Transportation Safety Board recommendations related to pipelines as they may pertain to the specific repair situation.

# SECTION 3—PREDEPARTURE PRACTICES

## 3.1 General

When piping revisions, replacements, or repairs are needed, the goal is to safely and efficiently complete the work. To ensure the safe and efficient completion of the job, the proper preplanning and groundwork must be accomplished before any work is done.

## 3.2 Preplan Procedures

The preplan should include the following steps:

a. Consult and comply with all applicable laws and regulations.
b. Initiate shut down if required.
c. Notify "ONE CALL" service if provided in the area or, if not, contact the owners of the buried facilities directly, giving notice of the work and allowing ample time for other buried facilities to be located and marked.
d. Evaluate damage and determine if a Safety-Related Condition Report should be filed with the Department of Transportation.
e. Develop a written workplan that includes the proper drain-down procedures and the proper drain-down equipment.
f. Obtain any required permits, including landowner, local government, other regulatory agencies, and hot work permits, if required.
g. Ensure all tools and safety equipment are in good working order.
h. Brief all line repair personnel on the proper safety procedures.
i. Notify personnel at the control center and discuss the line repair plans. Pipeline flow conditions, such as pressure and flow rate at the site of the repair, should be considered prior to the actual initiation of the repair. Emergency procedures should also be reviewed with the control center personnel.
j. Assure adequate fire protection equipment is available for emergencies.
k. Assure all communications equipment is in good working order and establish a procedure for the control center to monitor the job's progress.
l. Assure all communications and other electronic equipment is intrinsically safe or ensure that this equipment will be used a safe distance from the worksite.
m. Close, lock, and tag the valves connected to the line section to be repaired.
n. Open, lock, and tag the circuit breakers on any pumps and motor operated valves that could affect the safety of the job.

# SECTION 4—AREA ASSESSMENT

## 4.1 General

If a repair is required because of damage that has resulted in a petroleum product leak or the petroleum product has spilled during the repair, a hazard assessment must be done. This hazard assessment should result in a worksite safety plan. A hazard zone should be established by determining the amount of hydrocarbon vapors in the area. Information that is contained in the Material Safety Data Sheet must be consulted to determine the material's volatile and toxic characteristics. The initial monitoring of the area should be completed with full protective equipment as recommended by the Material Safety Data Sheet.

## 4.2 Area Assessment Procedures

An area assessment should include the following steps:

a. The area's flammability should be checked first with a combustible gas indicator. Areas above 10 percent of the lower explosive limit (LEL) should be declared hazardous, and access should be restricted. Areas above 50 percent of the LEL should not be entered.

b. Toxicity testing should be made with direct reading instruments that may include colorimetric tubes. This toxicity testing will determine the level of protective equipment required for the area.

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c. If a confined space is involved, such as a trench, always check for oxygen.
d. Personnel and equipment should not be permitted in the area until the hazard area has been defined.
e. Restrict equipment and other ignition sources from the flammable hazard zone.
f. All personnel who enter the toxic hazard zone must be equipped with proper protective gear.
g. If the hazard area extends into public areas, local authorities should be used to warn the public and to restrict access to the area. If public access is possible, suitable warnings should be posted.
h. The leaksite should be barricaded and marked to prevent the possibility of accidents and injuries. Spectators should not be permitted within the hazard area at any time.
i. Some leaks, according to their severity or hazards, may require compliance with the Occupational Safety and Health Administration's (OSHA) Hazwoper Standard (29 *Code of Federal Regulations* Part 1910.120).

# SECTION 5—EXCAVATION

## 5.1 General

All excavations deeper than 5 feet that personnel must enter must conform to the OSHA Construction Standard (29 *Code of Federal Regulations* Part 1926, Subpart P). The OSHA Construction Standard regulates the use of support systems, sloping and benching, and other systems of protection against excavation cave-ins. In addition, it regulates the means of access to and egress from excavations and employee exposure to vehicular traffic, falling loads, hazardous atmospheres, water accumulations, and unsafe structures in and adjacent to excavations. Excavations are defined in the Construction Standard to include trenches.

## 5.2 OSHA Construction Standard Checklist

To ensure that existing and predictable hazards are identified at the worksite, the following checklist is provided as an outline of the OSHA Construction Standard:

a. Is the excavation more than 5 feet in depth?
b. Is the excavation more than 20 feet in depth? (If yes, a professional engineer will be required to design a shoring system.)
c. Has a competent person (as defined by the OSHA standard) been designated?
d. Has the soil been categorized (stable rock, Type A, Type B, or Type C)?
e. Does the excavation require sloping, shoring, or shielding?
f. If excavation protection is required, which system will be used?
g. Have "ONE CALL" notifications been made or other means used to determine what underground utilities are in the area?
h. Has a stairway, ladder, ramp, or other safe means of egress been provided?
i. Has the excavation been tested for a hazardous atmosphere (flammability, oxygen content, toxicity)?
j. Has adequate protection been provided to protect employees from materials falling or rolling from the excavation face or from the surface in the vicinity of the excavation?
k. Has adequate protection been provided for water accumulation in the excavation?
l. Is emergency rescue equipment available at the jobsite?
m. Has stability of adjacent structures been considered as a result of the excavation?
n. Are daily inspections of the excavations being made?
o. Has a safe installation and removal procedure been developed to protect the employees who will be performing these tasks?

This checklist is designed only as a reminder; the details of the OSHA Construction Standard can be found in Subpart P of 29 *Code of Federal Regulations* Part 1926.

# SECTION 6—REPAIR PRACTICES

## 6.1 General Repair Procedures

If damage or imperfections are discovered on a pipeline, a decision to repair or not to repair will be made. When practical, an injurious pipe defect may be completely removed from the pipeline by replacing the portion of the pipe containing the defect. The new piece of pipe must meet at least the minimum design requirements of the pipeline. The length of the replacement piece should be a minimum of one half the diameter of the pipe but no shorter than 12 inches. No piece shall be installed closer than 6 inches to an existing weld. If pipe replacement is not practical, other methods of

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permanent repairs are covered in ASME B31.4. If repairs are warranted but not immediately practical, temporary repairs may be considered.

Temporary repairs (that is, clamps or mechanically-applied full encirclement sleeves) may be necessitated for operating purposes. The temporary repairs shall be made in a safe manner and in accordance with the manufacturer's instructions. As soon as practical, the temporary repairs shall be replaced in a permanent manner or welded permanently, if so designed. Derating (lowering) the line's maximum operating pressure may be required until permanent repairs are made, depending on the conditions and the design of the temporary repair. For particularly sensitive or critical locations, the operator may wish to consider leaving the repair excavation site open and maintaining personnel onsite to monitor the temporary repair while it is in operation until the permanent repair can be completed. ASME B31.4 specifically outlines recommendations that address these issues.

When pipe replacement or the clamping procedure is not feasible or sufficient to permit returning the line to service, hot tapping the line to provide a means of inserting plugs (stopples) to isolate the defective line section may effect a safe repair (see API Publication 2209). A bypass may be installed around the leak to permit continued operations while the defective section is drained and repaired. Consideration should be given to factors such as product characteristics, pipe condition, temperatures, and required differential pressures to ensure that the proper type of plug is used and that its holding capacity can adequately resist the pressure to be encountered. Only skilled personnel trained for this operation should install and operate tapping and plugging equipment, or qualified representatives of the manufacturer should be available to assist.

If it is feasible to remove the pipeline from service, repairs may be accomplished by hot tapping the line and installing a connection through which the pipeline's contents can be drained, vented, or flared, as in the case of a highly volatile liquids (HVL) line, to a place that is safe for disposal. After draining has been completed and pressure has been reduced to atmospheric pressure, the line may be cold cut and the defective section replaced. Disposal of contaminated soil and other material should be handled or disposed of in accordance with federal, state, and local requirements. When a portion of the line is to be removed, there are several methods of safely preparing the line for welding the new section into place. These methods include, but are not limited to, inert gas purging, mud plugging, ice plugging, and mechanical plugging. Each method has advantages and disadvantages based on the repair conditions. See API Publication 2209 for more details.

## 6.2 Safe Repair Steps

To ensure that the pipeline is repaired in a safe manner, the following steps and precautions should be taken:

a. Repair techniques used shall be in accordance with ASME B31.4.

b. Repairs shall be performed under qualified supervision by trained personnel aware of and familiar with the hazards to personnel, the public, and the environment.

c. Calculations should be performed to determine the amount of pipe movement that is permissible during the proposed repair. These calculations should consider the pipe metallurgy so that proper pipeline support can be provided during the repair work.

d. Requirements for supporting pipe both during and after construction should be determined to ensure that the pipe is not overstressed and to maximize safety.

e. All welding and testing shall be in accordance with ASME B31.4, API Recommended Practice 1104, and API Standard 1107. All welds shall be made by qualified welders using qualified procedures.

f. When the carrier pipe is being welded, the wall thickness shall be confirmed through the use of ultrasonic equipment or other suitable devices.

g. All rectifiers in the area should be turned off, locked, and tagged. This includes the company rectifiers, as well as the foreign pipeline rectifiers whose line is bonded to the line under repair or whose line passes in proximity to the line.

h. Because of the possibility of electrical currents on the pipeline, an electrical bond should be made across all proposed points of separation before the line is cut or a flange joint is separated. If replacement pipe is required, the pipe joint or joints should also be bonded. The bond should not be removed until repairs have been completed.

i. Line cuts, when required, should be made with mechanical cutters.

j. If welding is to be performed, all oils, products, and saturated earth should be removed both from within and around the excavation. It may be necessary to spread uncontaminated dirt around and on the bottom of the excavation.

k. The excavation and its surrounding area should be tested and continuously monitored with a combustible gas indicator or oxygen monitor, or both, to determine that the atmosphere is safe for such work.

l. Where vapor seals or plugs are used to prevent the escape of vapor from a pipeline, some positive method of venting or monitoring should be used to ensure against a pressure buildup in the line while hot work is in progress.

m. If oil, product, or vapors enter the excavation after hot work is started, the work should be halted immediately, and the oil, product, or vapor removed. The atmosphere should be retested prior to continuing the hot work.

n. Upon completion of the repair, necessary inspections, tests, and operating checks should be made before placing the line in service.

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# SECTION 7—POST-JOB PRACTICES

After a repair job is completed, there are post-job considerations that help ensure that the effectiveness of the preplanning and the on-the-job plan execution is not wasted or compromised by improper work completion practices. The following checklist provides some of those considerations:

a. After all repair work is completed and inspected, personnel at the control center should be notified that the line is ready for service.
b. All valves and circuit breakers should be untagged and operated in accordance with company procedures and control center instructions. All affected personnel should be notified.
c. After startup, the line repair should be visually monitored for leaks until normal operation is restored.
d. All rectifiers should be untagged and turned on.
e. The backfilling operation should be done in a careful manner so as not to damage any existing or new pipe coating and to affect good compaction.
f. Following the completion of a permanent repair, the site should be restored.
g. Contaminated soil and other materials should be handled or disposed of in accordance with local, state, and federal regulations.
h. All documentation should be completed, including but not limited to, the type of repair made and the materials used, the hydrotest and welding records, the weld locations, and any evidence of internal or external corrosion.
i. All DOT, 49 *Code of Federal Regulations* Part 195, and any other federal, state, or local reporting requirements should be completed.

# SECTION 8—SPECIAL CONSIDERATIONS FOR LIQUEFIED PETROLEUM GAS (LPG) OR OTHER HIGHLY VOLATILE LIQUIDS PIPELINES

## 8.1 Significant Characteristics and Their Related Problems

The preceding procedures are generally applicable to the repair of pipelines handling liquefied petroleum gas (LPG) or other highly volatile liquids, but personnel assigned to repair crews for pipelines should be well informed about the characteristics of the special materials that they may handle and the special problems that they may encounter if leaks occur. The most significant characteristics and their related problems are as follows:

a. The boiling points of liquefied petroleum gas materials are well below usual ambient temperatures; therefore, any liquid released as a result of a leak usually converts rapidly to vapor. Further, releases of such a liquid or vapor can create an explosive atmosphere over a large area.
b. Since the vapors of liquefied petroleum gas materials (like those of gasoline) are heavier than air and thus tend to remain close to the ground, the precautions outlined in the preceding sections are especially applicable to the potential hazards associated with liquefied petroleum gas leaks.
c. Vaporization of leaking liquefied petroleum gas may freeze the surrounding ground and pipeline appurtenances, and the danger exists that escaping gas could cause frostbite if it contacts exposed parts of the body.
d. The refrigerating effect of liquefied petroleum gas on the ground can also cause difficulties in excavation.
e. Since liquefied petroleum gases have substantially greater volatility than crude oil or gasoline, additional precautions may be required when leaks occur.

## 8.2 Precautions

Following the detection of a leak, the following precautions (not necessarily in the order shown) should be taken:

a. Eliminate all nearby ignition sources (especially those downwind of the leak) and evacuate adjacent areas that may be in danger.
b. Determine with a combustible gas indicator the extent of any explosive atmosphere in the area.
c. If conditions warrant, contact appropriate public officials for assistance in isolating the area, controlling traffic, evacuating nearby residential areas, and controlling spectators.
d. If explosive vapors are not accumulating to an extent that causes a hazard, consideration should be given to continue pumping until a less volatile product has replaced the liquefied petroleum gas at the point of leakage. If this procedure can be accomplished, the hazards associated with subsequent repairs to the line may be significantly reduced.

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